



Bharati Vidyapeeth's
College of Engineering, New Delhi

ECC-209

Analog Communication

Unit – 4



Bharati Vidyapeeth's College of Engineering, New Delhi

- Noise introduction
- Sources of noise
- Classification of noise
- Noise calculations (thermal noise)
- SNR
- Noise figure for cascaded amplifiers
- Noise Factor
- Effective input Noise Temperature.
- Superposition of noises

1. Introduction-Noise

- In general the term NOISE is used to describe an unwanted signal which affects (corrupts) a wanted signal.
- Noise in electrical terms may be defined as any unwanted introduction of energy tending to interfere with the proper reception and reproduction of transmitted signals.
- In general NOISE may be predictable or unpredictable (random) in nature.
- The predictable noise can be estimated and eliminated by proper design. Interference arises for example, from other communication systems (cross talk), 50 Hz supplies (hum) and harmonics, switched mode power supplies, thyristor circuits, ignition (car spark plugs) motors ... etc.
- Unpredictable noise varies randomly with time and no control over this noise.

1. Introduction-Noise

➤ Identification of the message signal at the receiver depends upon the amount of noise present in message signal during the process of communication. The amount of noise power present in the received signal reduced the power level of desired signal.

➤ These unwanted signals arise from a variety of sources which may be considered in one of two main categories:-

1.EXTERNAL NOISE :Noise created outside the receiver

2.INTERNAL NOISE :Noise created within the receiver itself.

It is created by active and passive elements present within the communication circuit itself. Fluctuation noise is caused by spontaneous fluctuation in the physical system. Such as thermal motion of the free electron inside a resistor, known as brownian motion .the random emission of electron in vacuum tube. random diffusion of electron and holes in a semiconductor.

Sources of noise

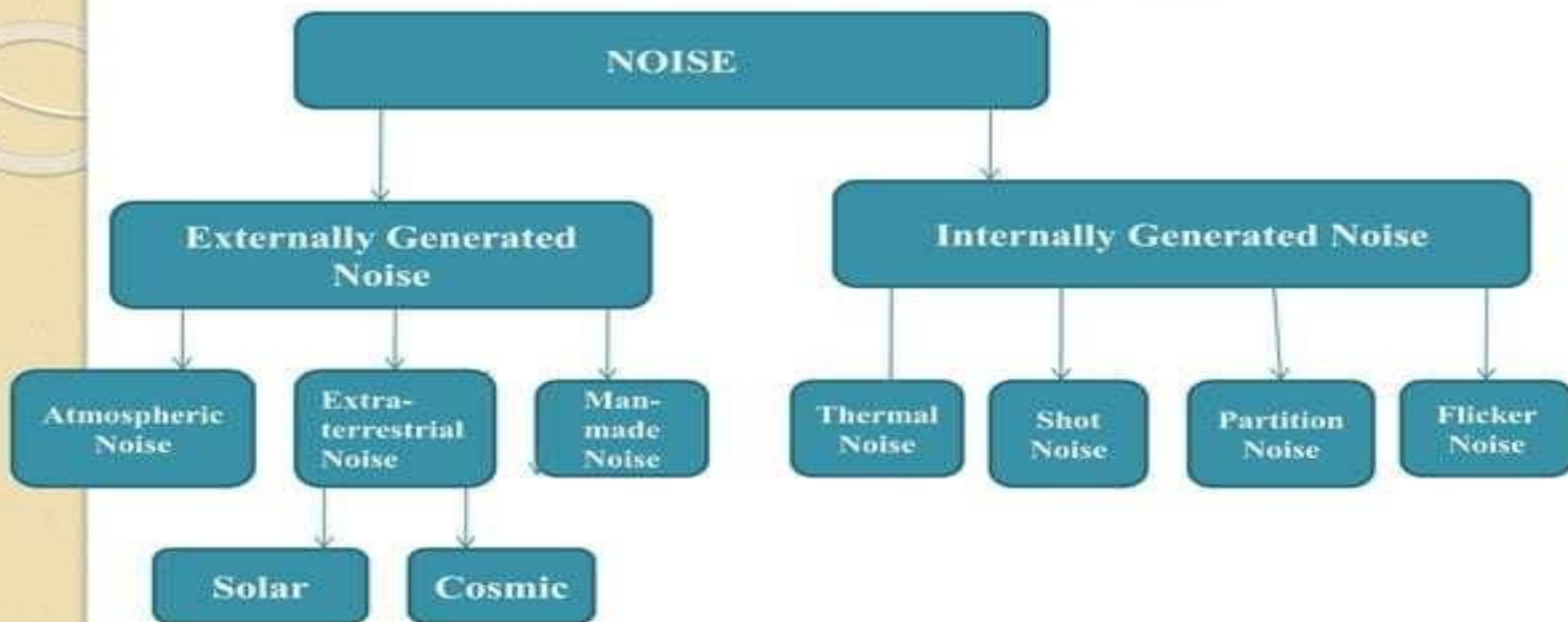
➤ External

- Atmospheric
- Industrial(Man made Noise)
- Extra-terrestrial
 - Solar noise
 - Cosmic noise

➤ Internal

- Thermal Noise
- Shot Noise
- Partition Noise
- Flicker Noise

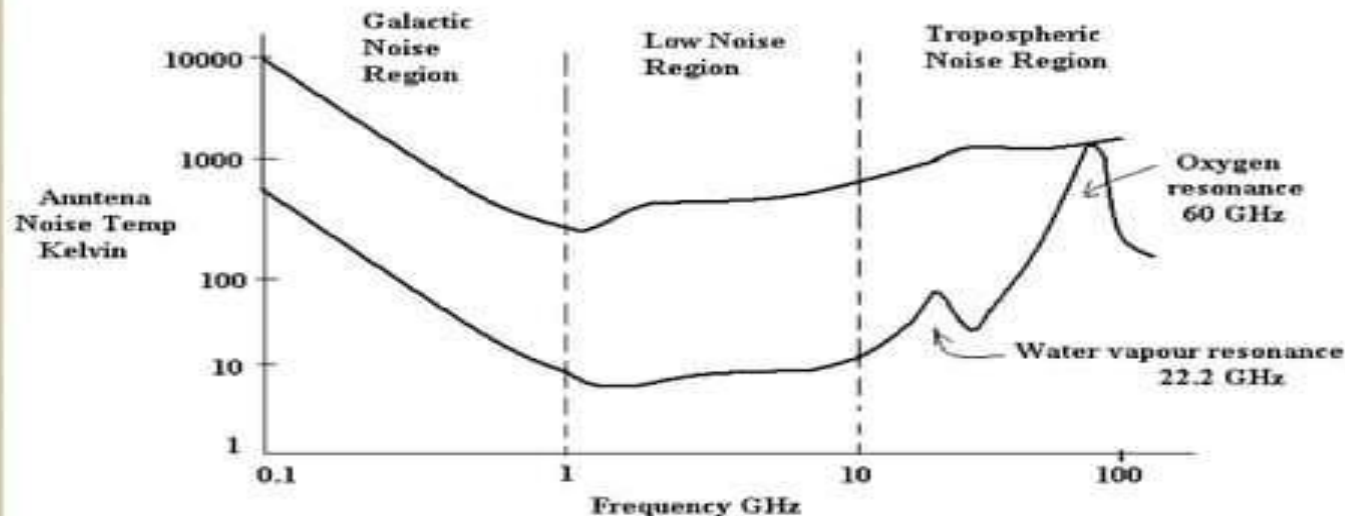
Sources of noise



External Noise

Natural Noise

Naturally occurring external noise sources include atmosphere disturbance (e.g. electric storms, lighting, ionospheric effect etc), so called 'Sky Noise' or Cosmic noise which includes noise from galaxy, solar noise and 'hot spot' due to oxygen and water vapour resonance in the earth's atmosphere.



1. Atmospheric Noise

Atmospheric noise also known as static noise. It is caused by naturally occurring disturbances in the earth's atmosphere

SOURCES

Static is caused by lightning discharges in thunderstorms and other natural electric disturbances occurring in the atmosphere.

Nature and Form

It comes in the form of amplitude modulated impulses. Such impulse processes are random and spread over the whole of the RF spectrum used for broadcasting. It consists of spurious radio signals with many frequency components.

Nature and Form

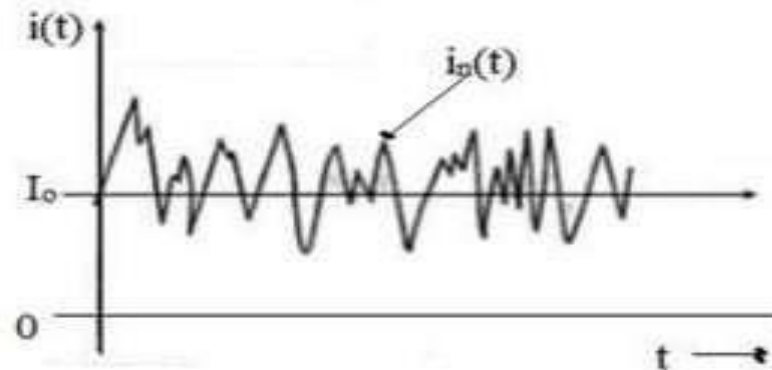
- It is propagated in the same way as ordinary radio waves of the same frequency.
- Any radio station will therefore receive static from thunderstorms both local and distant.
- It affects radio more than it affects television. The reason, field strength is inversely proportional to frequency.
 - At 30MHz and above atmospheric noise is less severe for two reasons:
 - Higher frequencies are limited to line of sight propagation, i.e., less than 80 kilometres or so on.
 - Very little of this noise is generated in the VHF range and above.

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1. Shot Noise

- Shot noise appears in active devices due to the random behavior of charge carriers (electrons and holes).
- In electron tubes, shot noise is generated due to the random emission of electrons from cathodes; in semiconductor devices, it is caused due to the random diffusion of minority carriers.
- Current in electron devices (tubes or solid state) flows in the form of discrete pulses, every time a charge carrier moves from one point to the other (e.g., cathode to plate). Therefore, although the current appears to be continuous is still a discrete phenomena. The nature of current variation with time is shown in Figure



2. Internal Noise

This is the noise generated by any of the active or passive devices found in the receiver. This type of noise is so known as fluctuation noise.

This type of noise is random and difficult to treat on an individual basis but can be described statistically.

Random noise power is proportional to the bandwidth over which it is measured .

It is caused by spontaneous fluctuations in physical system.

Examples of such fluctuations are;

(a) thermal motion of the free electrons inside a resistor, known as Brownian motion which is random in nature;

(b) the random emission of electrons in vacuum tubes and the random diffusion of electrons and holes in a semiconductor.

The fluctuation is very significant, and will be treated in greater detail.

The two important types of fluctuation noise are :

(i) Shot noise (ii) thermal noise.

3. Extraterrestrial-Noise (ii) **Cosmic noise**

Distant stars also radiate noise in much the same way as the sun. The noise received from them is called black body noise. Noise also comes from distant galaxies in much the same way as they come from the milky way.

Extraterrestrial noise is observable at frequencies in the range from about 8MHz to 1.43GHz.

Apart from man made noise it is strongest component over the range of 20 to 120MHz.

Not much of it below 20MHz penetrates below the ionosphere

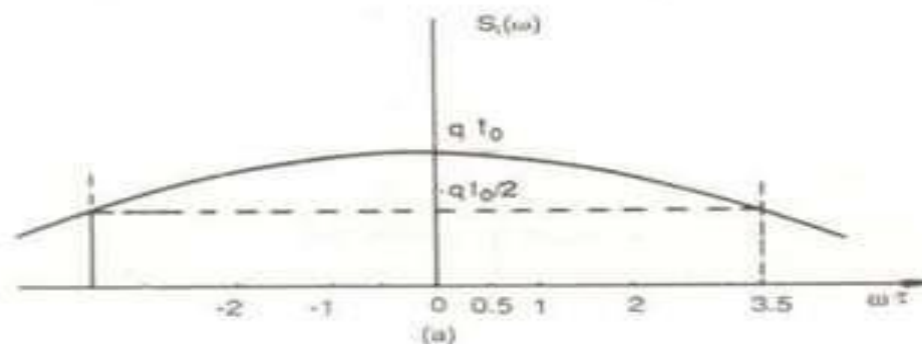
1. Shot Noise.....

- The current fluctuates about a mean value I_o . This current $i_n(t)$ which wiggles around the mean value is known as shot noise. The wiggling nature of the current is not visualized by normal instruments, and normally we think that the current is a constant equal to I_o . The wiggling nature of the current can be observed in a fast sweep oscilloscope. The total current $i(t)$ may be expressed as current $i(t) = I_o + i_n(t)$,
• where I_o is the constant (mean), and $i_n(t)$ is the fluctuating (noise) current. **Power Density Spectrum of Shot Noise In Diodes**
- The time varying component $i_n(t)$ of the current $i(t)$ specified by
- Eq. $i(t) = I_o + i_n(t)$ is random in nature, and it cannot be expressed as a function of time, i.e., it is an indeterministic function.
- However, this indeterministic stationary random function $i_n(t)$ can be specified by its power density spectrum. The number of electrons contributing to the random stationary current $i_n(t)$ is large.
- Assuming that the electrons do not interact with each other during their movement, or emission (e.g., temperature limited diode current), the process may be considered statistically independent. According to central limit theorem, such process has a Gaussian distribution. Hence, shot noise is Gaussian-distributed with zero mean.

1. Shot Noise

The total diode current may be taken as the sum of the current pulses, each pulse being formed by the transit of an electron from cathode to anode. It can be seen that for all practical purposes the power density spectrum of the statistically independent non interacting random noise current, is given by

$S_i(\omega) = qI_0$ where q is the electronic charge and I_0 , is the mean value of the current in amperes, The power density spectrum is frequency independent, This type of frequency independence is only up to a frequency range decided by the transit time of an electron to reach from anode to cathode. Beyond this frequency range, the power density varies with frequency as shown in Fig. 4.2.2a.



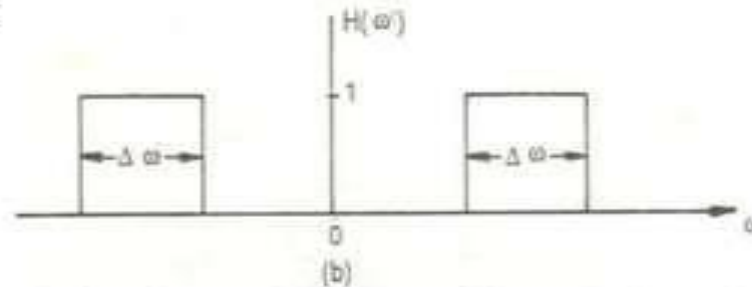
1. Shot Noise

The transit time of an electron, in a diode, depends on anode voltage V and is given as $\tau = 3.36 \times (d/\sqrt{V}) \mu\text{sec}$ where d is spacing between anode and cathode. For instance, in a diode with $d = 2\text{mm}$ and $V = 40$ volts, we have $\tau = 10^{-3} \mu\text{sec}$. In Fig. 4.222a the power density curve may be considered flat close to the origin, i.e. $|\omega\tau| \leq 0.5$. Therefore $S_i(\omega)$ can be considered constant up to $|\omega\tau| \leq 0.5$. For $\tau = 10^{-3} \mu\text{sec}$, the maximum frequency up to which power density remains constant is given by $\omega = 0.5 \times 10^9 = 5 \times 10^8$ rad/sec.

This is equivalent to a linear frequency $f = \omega/2\pi \approx 80$ MHz. Therefore for all practical purposes, the $S_i(\omega)$ may be considered to be frequency independent below 100 MHz. This frequency limit covers the frequency range of most of the practical communication systems, except UHF and microwave.

1. Shot Noise.....

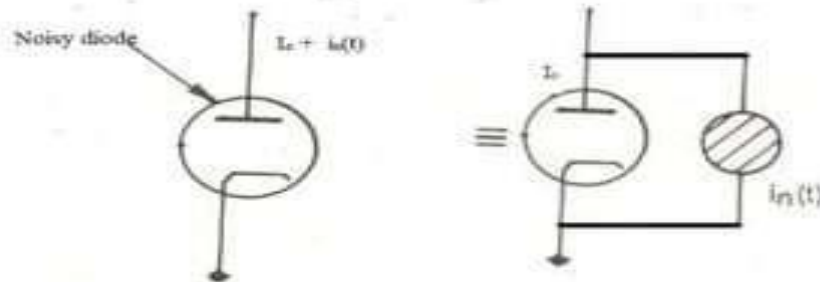
Schottky Formula The mean square value (average power of the randomly fluctuating noise current will be $i_n^2 = 2qI_0 \Delta f$ (4.2.3) where $2\Delta f$ is the bandwidth (including negative frequency) of the measuring system involved, as shown in Figure b. of course below 100 MHz). The Schottky formula.



Eq. $S_i(\omega) = qI_0$ has been developed assuming that electrons contributing the diode current do not interact with each other, as in the case of a temperature limited region of a thermionic diode. There may be cases where electrons contributing the diode current interact with each other as in a space-charge limited region of a thermionic diode.

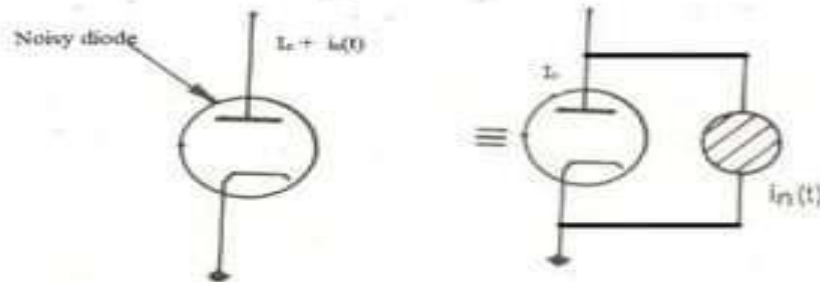
1. Shot Noise.....

The In such cases, power density spectrum is given by $S_i(\omega) = \alpha q I_0$. (4.2.4) where α is a smoothing constant, and ranges between 0.01 to 1. The space-charge has a smoothing effect and depends on the tendency of interacting electrons to smooth out and yield a constant current. The more is the smoothing effect, the greater is the value of α . It is expressed as $\alpha = 1.288 k T_c g_d / q I_0$ where T_c is cathode temperature in degrees Kelvin; k is the Boltzmann constant ($k = 1.38 \times 10^{-23}$ Joules per degree Kelvin), and is the dynamic conductance of the diode. Substituting this value of α , Eq. $S_i(\omega) = \alpha q I_0$ 4.2.4 becomes $S_i(\omega) = 1.288 k T_c g_d / I_0$. An equivalent circuit of a noisy diode in terms of a noiseless diode is shown in Fig. 4.



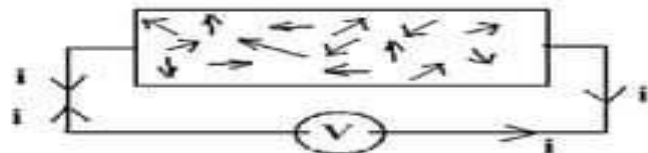
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2. Thermal Noise (Johnson Noise/Resistor Noise)

- The noise arising due to random motion of free charged particles (usually electron) in a conducting medium, such as a resistor, is called resistor noise. This is also known as Johnson noise after J.B. Johnson who investigated this type of noise in conductors.
- The random agitation is a universal phenomenon at atomic levels and is caused by the energy supplied through flow of heat.
- The intensity of random motion is proportional to thermal (heat) energy supplied (ie, temperature), and is zero at a temperature of absolute zero. This noise is also known as thermal noise. The path of the electron motion is random of the collisions with lattice structure. The net motion of all the electrons gives rise to an electric current to flow through the resistor, causing the noise

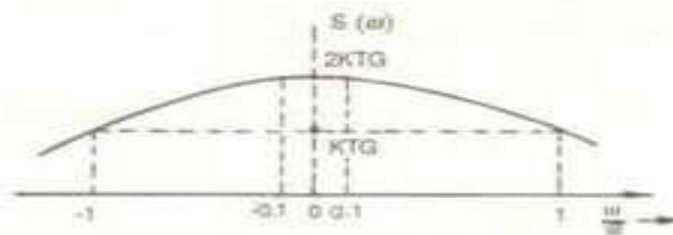


- This type of noise is generated by all resistances (e.g. a resistor, semiconductor, the resistance of a resonant circuit, i.e. the real part of the impedance, cable etc).

2. Thermal Noise (Johnson Noise/Resistor Noise)

Power Density Spectrum of Resistor Noise

- The free electrons contributing to resistor noise are large in number.
- If their random motion is assumed to be statistically independent, then the central limit theorem predicts the resistor noise to be, gaussian distributed with a zero mean.
- It can be shown that the power density spectrum of the current contributing the Thermal noise is given by $S_i(\omega) = 2KTG/[1 + (\omega/\alpha)^2]$
- where T is ambient temperature in degree Kelvin, G is the conductance of the resistor in mhos, k is the Boltzman constant, α is the average number of collisions per second per electron.
- The variation of power density spectrum with frequency is shown in Fig.43.1



2. Thermal Noise (Johnson Noise/Resistor Noise)

Power Density Spectrum of Resistor Noise

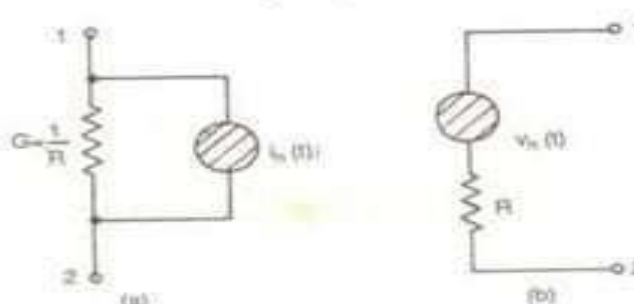
- It is obvious from the figure that the spectrum may be conducted to be flat for $(\omega/\alpha) \leq 0.01$ so.
- The power density spectrum $S_i(\omega)$ for this range of frequency is nearly constant and given by $S_i(\omega) = 2KTG$.
- The value of α is of the order of 10^{14} and hence the frequency range corresponding to of 10^{13} Hz.
- Therefore, the frequency independent expression of $S_i(\omega)$ gives by Eq. $S_i(\omega) = 2KTG$ holds up to frequency range of 10^{13} Hz.
- This range covers almost all the practical applications in communication systems.
- Hence for all practical purposes, the power density spectrum $S_i(\omega)$ is considered to be independent of frequency

2. Thermal Noise (Johnson Noise/Resistor Noise)

Equivalent Circuit of a Noise Resistor

➤ A noisy resistor R can be represented by noiseless conductance G in parallel with thermal noise current source $i_n(t)$ as shown in Fig (a). The Thevenin equivalent of Fig. a is shown in Fig. b,

➤



➤ Noiseless Resistor (a) With Noise Current Source (b) With Noise Voltage Source

➤ which represents the noiseless resistor R in series with a thermal noise voltage source $v_n(t)$. Current $i_n(t)$ and voltage $v_n(t)$ are related as $v_n(t) = R i_n(t)$.

➤ Now, since the power density spectrum is a function of the square of voltage or current, the relation between the power density spectrum $S_i(\omega)$ of the current source $i_n(t)$ and $S_v(\omega)$ of the voltage source is given as

$$➤ S_v(\omega) = R^2 S_i(\omega) = R^2(2kTG) = R^2(2kT/R) = 2kTR = S_v(\omega)$$

2. Thermal Noise (Johnson Noise/Resistor Noise)

Power of Thermal Noise Voltage

➤ The power density spectrum $S_v(\omega)$ of a thermal noise voltage $v_n(t)$ is independent of frequency.

➤ Since power density spectrum is the power per unit bandwidth, noise power increases with an increase in bandwidth and becomes infinite as the bandwidth tends to infinity. This is obvious from the relation for noise power P_n given by

➤

$$P_n = (1/2\pi) \int_{-\infty}^{\infty} S_v(\omega) d\omega \quad P_n = (1/2\pi) \int_{-\infty}^{\infty} 2kTR d\omega$$

➤ The integral becomes infinite when integrated over an infinite bandwidth. However, for a finite bandwidth of $2\Delta f$ ($-\Delta f$ to Δf) the noise power (m.s. value) is given by $P_n = S_v(\omega) \cdot 2\Delta f = 4kTR\Delta f$

➤ Since, the power of a signal is the same as its mean square value

$$P_n = v_n^2 = 4kTR\Delta f.$$

➤ The corresponding rms value is given by $P_n = v_n^2 = 4kTR\Delta f$. Note that Δf is one sided (positive) bandwidth. The thermal noise power contribution is limited only by the bandwidth of the circuit.

3. White Noise

- White light contains all colour frequencies.
- In the same way, white noise, too, contains all frequencies in equal amount.
- The power density spectrum of a white noise is independent of frequency: which means it contains all the frequency components in equal amount .
- When the probability of occurrence of a white noise level is specified by a Gaussian distribution function, it is known as White Gaussian noise.
- The power density spectrum of that noise is independent of the operating frequencies which is given by $S_v(\omega) = 2kTR$.
- Hence, shot noise and thermal noise may be considered as white Gaussian noise for all practical purposes.
- The power density of white noise is $S_w(\omega) = \eta_0 / 2$
white noise contains all Frequency components, but the phase relationship of the components is random, whereas the Delta function has all frequency components with equal magnitude, and the same relative phase.
- The inverse Fourier transform of white noise is specified by autocorrelation function wherein phase relationship has no significance Thus the autocorrelation function of a white noise is a Delta function.

3. White Noise

This is shown in Figure below:



Fig (a) autocorrelation (b) Power spectrum of a white noise

- where it is obvious that Delta function and the power density spectrum of white noise are a Fourier transform pair.
- White noise has infinite power and is not physically realizable. But, its concept is helpful in convenient mathematical analysis of systems.
- The autocorrelation is zero for $\tau \neq 0$; i.e., any two samples of white noise are uncorrelated, and also if white noise is Gaussian, the two samples are statistically independent
-

3. Partition Noise

- **Definition:** Partition noise occurs wherever current has to divide between two or more paths; and partition noise results due to the random fluctuations in the division of current.
- In a bipolar junction transistor, there is noise due to the random motion of the carriers crossing emitter-base and base-collector junctions, and to random recombination of holes and electrons in the base. As the emitter is divided into base and collector current, there is a partition effect arising from the random fluctuations in the division of current between the collector and the base.
- It is observed that a transistor does not generate white noise, except over a midband region. Also, the noise generated depends upon the quiescent conditions and the source resistance. Hence, in specifying the noise in transistor, the center frequency, the operating point, and source resistance must be specified.
- In a p-n junction diode, there is no division of current; and hence, if all other factors are equal, then a diode is less noisy than a transistor. It is for this reason that in microwave receivers, where bandwidth is large, diode mixers are used to minimize noise. For low noise microwave amplification, zero gate current gallium arsenide field-effect transistors are specially developed

3. Partition Noise

- The main source of noise in the FET is the thermal noise of the conducting channel. Gate leakage current, having small fluctuations with time, give rise to shot noise. It should be noted that, unlike the bipolar junction transistor
- the noise figure of the FET is essentially independent of the quiescent operating point [V_{DSQ} , I_{DQ}]

4. Flicker Noise or Low Frequency

° Active devices, integrated circuit, diodes, transistors etc also exhibits a low frequency noise, which is frequency dependent (i.e. non uniform) known as flicker noise or 'one – over – f ' noise.

Definition: In semiconductor devices, flicker noise arises due to fluctuations in carrier density . The conductivity of the semiconducting material depends on carrier density .As carrier density fluctuates, conductivity will also fluctuate. When the direct semiconductor, fluctuating voltage drop is produced, which the flicker-noise voltage.

The mean square value of flicker noise voltage is proportional to the square of the

direct current flowing through the semiconducting material.

The flicker noise is appears below frequencies of few kHz. The spectrum density of noise increases as frequency decreases, hence it is sometimes referred to as $(1/f)$ noise, i.e. noise varying inversely with frequency

4. High Frequency or Transit Time Noise

- In semiconductor devices, the transit time is the time taken by the carriers to cross a junction. The periodic time of the signal is equal to reciprocal of signal frequency. When the signal frequency is high, periodic time becomes very small and hence may be comparable to transit time of carriers. In such situation, some of the carriers may diffuse back to the source, i.e. emitter. Due to this, conductance component of input admittance increases with frequency. This conductance has associated with a noise current generator [$I_n^2 = 4GkT\Delta f$]. Since the conductance increases with frequency, the noise spectrum density increases at high frequencies

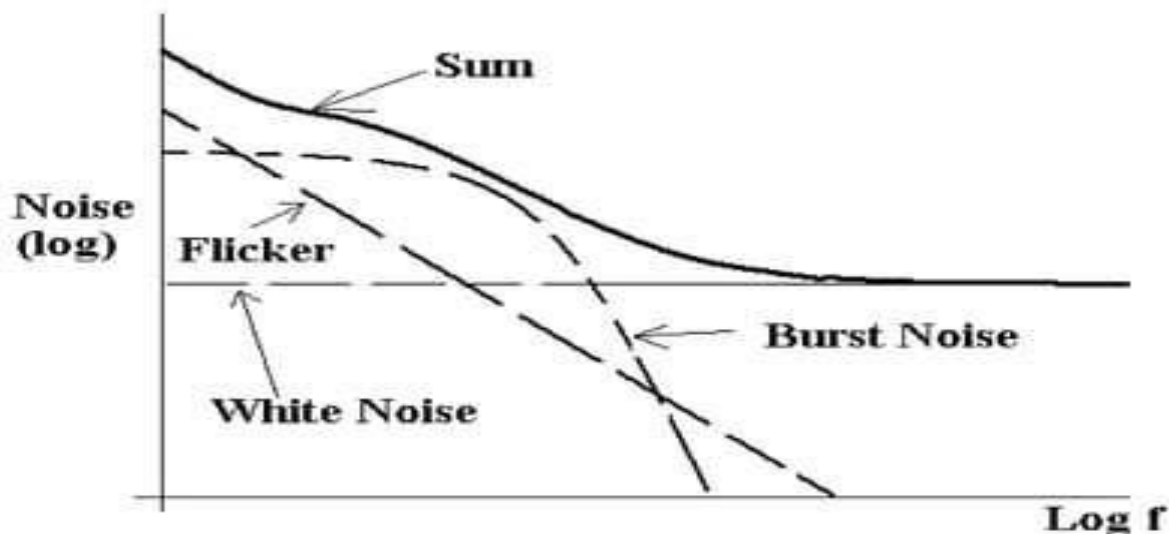
6. Burst Noise or Popcorn Noise

- Definition: The burst noise appears as a series of bursts at two or more levels. It appears in bipolar transistors and is of low frequency nature.
- The burst noise produces popping sounds in an audio system. Hence it is also called popcorn noise.
- The spectral density of burst noise increase as the frequency decrease
- Such semiconductors which produce burst(or) popcorn noise has a spectral density proportional to

5. Excess Resistor Noise

Thermal noise in resistors does not vary with frequency, as previously noted, by many resistors also generates as additional frequency dependent noise referred to as excess noise.

7. General Comments



For frequencies below a few KHz (low frequency systems), flicker and popcorn noise are the most significant, but these may be ignored at higher frequencies where 'white' noise predominates.

Signal to Noise

The signal to noise ratio is given by

$$\frac{S}{N} = \frac{\text{Signal Power}}{\text{Noise Power}}$$

The signal to noise in dB is expressed by

$$\left(\frac{S}{N}\right)_{dB} = 10 \log_{10} \left(\frac{S}{N}\right)$$

$$\left(\frac{S}{N}\right)_{dB} = S_{dBm} - N_{dBm} \text{ for } S \text{ and } N \text{ measured in mW.}$$

Noise Factor- Noise Figure

Consider the network shown below,



Noise Factor- Noise Figure (Cont'd)

- The amount of noise added by the network is embodied in the Noise Factor F , which is defined by

$$\text{Noise factor } F = \frac{\left(\frac{S}{N}\right)_{IN}}{\left(\frac{S}{N}\right)_{OUT}}$$

- F equals to 1 for noiseless network and in general $F > 1$.
The noise figure in the noise factor quoted in dB
i.e. Noise Figure F dB = $10 \log_{10} F$ $F \geq 0$ dB

- The noise figure / factor is the measure of how much a network degrades the $(S/N)_{IN}$, the lower the value of F , the better the network.

Noise Factor- Noise Figure (Cont'd)

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System Noise Temperature

Since $T_e = (F-1)T_s$, i.e. $F = 1 + \frac{T_e}{T_s}$

Then

$$F_{sys} = 1 + \frac{T_{e sys}}{T_s}$$

where $T_{e sys}$ is the equivalent Noise temperature of the system
and T_s is the noise temperature of the source

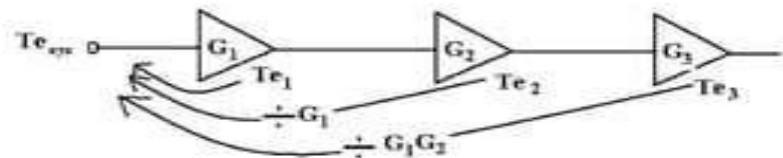
and

$$\left(1 + \frac{T_{e sys}}{T_s}\right) = \left(1 + \frac{T_{e1}}{T_s}\right) + \frac{\left(1 + \frac{T_{e2}}{T_s} - 1\right)}{G_1} + \dots etc$$

$$i.e. from F_{sys} = F_1 + \frac{(F_2 - 1)}{G_1} + \dots etc$$

which gives

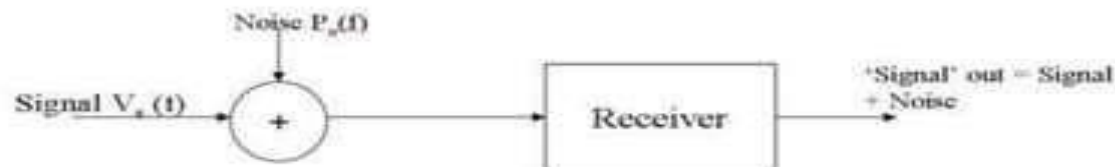
$$T_{e sys} = T_{e1} + \frac{T_{e2}}{G_1} + \frac{T_{e3}}{G_1 G_2} + \frac{T_{e4}}{G_1 G_2 G_3} + \dots$$



Additive White Gaussian Noise

Additive

Noise is usually additive in that it adds to the information bearing signal. A model of the received signal with additive noise is shown below



White



$$\text{White noise } = p_o(f) = \text{Constant}$$

Gaussian

We generally assume that noise voltage amplitudes have a Gaussian or Normal distribution.